

I Hybrid Normalization

Rescales Each Channel So No Instrument Dominates

WHY Sources Differ By Orders Of Magnitude In Scale

COMPARED WITH Z-score, Min-Max, Log

USED **Hybrid**

II Borderline Cleaning

Removes NSEP Samples That Sit On The SEP Boundary

WHY Ambiguous NSEP Samples Mislead The Classifier

COMPARED WITH ENN, NearMiss-3

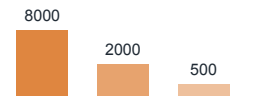
USED **Tomek Links**

III Random Under-Sampling & Over-Sampling

RUS

Randomly Drops NSEP Samples; All SEP Kept

LEVELS TRIED



WHY

We Use RUS To Reduce Imbalance Rate

OS

Generates Synthetic SEP Sequences From Real Ones

WHY

Only 118 Real SEP Sequences To Train On

FAMILIES TRIED

SMOTE, ADASYN

TimeGAN, Diffusion

APPLIED AT

Each NSEP Level

IV Training

Trains Four Architectures On Every Preprocessed Set

WHY Checks Gains Hold Across Model Families, Not Just One

MODELS USED SVM (catch22), GRU, PatchTST, InceptionTime

BEST

GRU